

Gaussian Approximation for Matching-based Average Treatment Effect (ATE) Estimators

Joint with K. Balasubramanian, W. Polonik and Z. Shi

Chinmoy Bhattacharjee

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Setup: The Causal Inference Problem

We observe i.i.d. triplets $(X_i, Y_i, D_i)_{i=1}^n \in \mathbb{X} \times \mathbb{R} \times \{0, 1\}$.

- X_i : Covariates supported on a compact set $\mathbb{X} \subset \mathbb{R}^m$.
- D_i : Treatment assignments, possibly dependent on X_i (1 for treated, 0 for control).
- $Y_i(0), Y_i(1)$: Potential outcomes under control and treatment. [Rubin '74]
- $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$: Observed outcomes.

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Goal: Estimate the Average Treatment Effect (ATE) across the population, defined by

$$\tau := \mathbb{E}[Y(1) - Y(0)].$$

Nearest-Neighbor Imputation

To estimate τ , impute the missing PO via **M-NN matching**. [[Abadie and Imbens '06](#)]

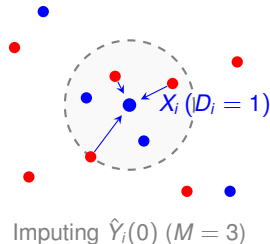
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Let $\mathcal{J}_M(i)$ denote the index set of the M -NNs of X_i among the units in the **opposite** treatment group. We estimate the missing POs as

$$\hat{Y}_i(0) = \begin{cases} Y_i, & \text{if } D_i = 0, \\ \frac{1}{M} \sum_{j \in \mathcal{J}_M(i)} Y_j, & \text{if } D_i = 1. \end{cases}$$

$$\hat{Y}_i(1) = \begin{cases} \frac{1}{M} \sum_{j \in \mathcal{J}_M(i)} Y_j, & \text{if } D_i = 0, \\ Y_i, & \text{if } D_i = 1. \end{cases}$$



The Unadjusted Estimator

Matching-based ATE estimator:

$$\hat{\tau}_M := \frac{1}{n} \sum_{i=1}^n \hat{Y}_i(1) - \frac{1}{n} \sum_{i=1}^n \hat{Y}_i(0).$$

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The weighting perspective:

Let $K_M^{D_i}(i) = \#$ times unit i is used as an M -NN match for the opposite group. Then

$$\hat{\tau}_M = \frac{1}{n} \sum_{i=1}^n (2D_i - 1) \left(1 + \frac{K_M^{D_i}(i)}{M} \right) Y_i.$$

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➤ $\hat{\tau}_M$ suffers from an asymptotically non-negligible bias for $m > 1$. [[Abadie & Imbens '06](#)]

Bias Correction & CLT

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Let $\hat{\mu}_\omega(x)$ estimate $\mu_\omega(x) = \mathbb{E}[Y \mid X = x, D = \omega]$ with residuals $\hat{R}_i := Y_i - \hat{\mu}_{D_i}(X_i)$.

$$\hat{\tau}_M^{bc} := \underbrace{\frac{1}{n} \sum_{i=1}^n (\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i))}_{\text{regression-based ATE estimator}} + \frac{1}{n} \sum_{i=1}^n (2D_i - 1) \left(1 + \frac{K_M^{D_i}(i)}{M} \right) \hat{R}_i.$$

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An asymptotic CLT for $\hat{\tau}_M^{bc}$ was shown in [Abadie & Imbens '11] (fixed M) and [Lin, Ding & Han '23] ($M \rightarrow \infty$).

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An asymptotic CLT for $\hat{\tau}_M^{bc}$ was shown in [Abadie & Imbens '11] (fixed M) and [Lin, Ding & Han '23] ($M \rightarrow \infty$). We want to quantify dependence M , $\mathbb{P}(D = 1 \mid X = x)$ and m .

Assumption A: Data Distribution

➤ **Bounded density:** $\mathbb{X} \subset \mathbb{R}^m$ is compact and convex. The covariate density g satisfies

$$0 < g_{\min} \leq g(x) \leq g_{\max} < \infty.$$

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Compared to [Lin, Ding & Han '23], we relax Lipschitz continuity to Hölder continuity.

Assumption B: Regression Models & Errors

- **Smoothness:** The true regression functions $\mu_\omega(x) = \mathbb{E}[Y \mid X = x, D = \omega]$ are continuously differentiable up to order $k := \lfloor m/2 \rfloor + 1$.
- **Estimator convergence:** The regression estimators $\hat{\mu}_\omega(x)$ and their mixed derivatives up to order $\lfloor m/2 \rfloor$ converge to the true functions at rates $n^{-\gamma_l}$.

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- **Error moments:** The errors have bounded $(4 + p)$ conditional moments for $p > 0$:

$$\mathbb{E}[|Y(\omega) - \mu_\omega(X)|^{4+p} \mid X = x, D = \omega] \leq M_{u,p} \quad \forall x, \omega.$$

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Compared to [Lin, Ding & Han '23], we require $(4 + p)$ moments (as opposed to $2 + p$).

Main Non-Asymptotic Bound

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Theorem (Shi, B., Balasubramanian, Polonik '26)

Let Assumptions A and B hold. If $M \lesssim n\eta$ and $n\eta^2 \gtrsim 1$, there exists $C > 0$ such that

$$d_K \left(\frac{\sqrt{n}(\hat{\tau}_M^{bc} - \tau)}{\sigma}, \mathcal{N}(0, 1) \right) \leq C(B_1 + B_2 + B_3).$$

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The limiting variance σ^2 is given by

$$\sigma^2 := \underbrace{\text{Var}(\mu_1(X) - \mu_0(X))}_{\text{Covariate Variability}} + \underbrace{\mathbb{E} \left[\frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1 - e(X)} \right]}_{\text{Outcome / Residual Variability}},$$

where $e(x) := \mathbb{P}(D = 1 | X = x)$ and $\sigma_\omega^2(x) := \mathbb{E}[(Y(\omega) - \mu_\omega(X))^2 | X = x]$.

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Corollary (Explicit rates) If additionally $\eta \geq \eta_0 > 0$, and $\log n \ll M \ll n/\log n$, then

- $B_1 \approx M^{\frac{40}{8+p}} n^{-1/2}$ (Stabilization Error)
- $B_2 \approx M^{\frac{k}{2m}} n^{-\frac{k}{2m} + \frac{1}{4}} + \max_{l \in [k-1]} \left(n^{-\frac{\gamma_l}{2} - \frac{l}{2m} + \frac{1}{4}} M^{\frac{l}{2m}} \right)$ (Residual Bias Error)
- $B_3 \approx \left(\frac{M}{n} \right)^{\frac{a}{2m}} + M^{-1/2} + n^{-1/3}$ (Variance Estimation Error).

Proof Idea I: Decomposition as sum of scores

$$\hat{\tau}_M^{bc} := \frac{1}{n} \sum_{i=1}^n (\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)) + \frac{1}{n} \sum_{i=1}^n (2D_i - 1) \left(1 + \frac{K_M^{D_i}(i)}{M} \right) (Y_i - \hat{\mu}_{D_i}(X_i)).$$

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Expanding around μ_ω and $\epsilon_i := Y_i - \mu_{D_i}(X_i)$, the estimator neatly decomposes as $\hat{\tau}_M^{bc} = E_n + \text{Residual Bias}$, where

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By regrouping, E_n can be expressed as a sum of **locally dependent** score functions of the Binomial process $\mathcal{X}_n := (X_i, Y_i, D_i)_{i=1}^n$:

$$E_n = \sum_{j=1}^n \xi_n((X_j, Y_j, D_j), \mathcal{X}_n).$$

Proof Idea II: Geometric Localization via Stabilization

- **Radius of stabilization:** The score $\xi_n((X_j, Y_j, D_j), \mathcal{X}_n)$ for unit j depends only on its local neighborhood of radius

$R_n((X_j, Y_j, D_j), \tilde{\mathcal{X}}_n) = \text{M-NN distance from } X_j \text{ in the opposite treatment group.}$

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- **Local dependence:** This rapid decay guarantees that the interactions are localized. Non-asymptotic bounds follow from [Lachièze-Rey, Schulte & Yukich '19].

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Multiplier Bootstrap: We resample the residuals $\hat{R}_i = Y_i - \hat{\mu}_{D_i}(X_i)$ using independent multipliers $W_i \sim \mathcal{N}(1, 1)$, and construct bootstrap outcomes $Y_i^* := \hat{\mu}_{D_i}(X_i) + W_i \hat{R}_i$. Further, we use independent $V_i \sim \mathcal{N}(0, 1)$ to perturb the regression predictions.

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Bootstrap Validity (Shi, Bhattacharjee, Balasubramanian, Polonik '26)

Conditional on the data $\mathcal{X}_n$, with high probability, we have

$$d_K\left(\sqrt{n}(\hat{\tau}_M^{boot} - \hat{\tau}_M^{bc}) \mid \mathcal{X}_n, \sqrt{n}(\hat{\tau}_M^{bc} - \tau)\right) \leq C\left(B_1 + B_2 + B_3 + \mathcal{E}(\mu, \hat{\mu})\right).$$

Proof Idea: Bootstrap Validity via Triangle Inequality

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$$d := d_K\left(\sqrt{n}(\hat{\tau}_M^{boot} - \hat{\tau}_M^{bc}) \mid \mathcal{X}_n, \sqrt{n}(\hat{\tau}_M^{bc} - \tau)\right)$$

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➤ Insert the true asymptotic Gaussian limit $\mathcal{N}(0, \sigma^2)$ as a mathematical bridge:

$$d \leq \underbrace{d_K\left(\text{Boot} \mid \mathcal{X}_n, \mathcal{N}(0, \sigma^2)\right)}_{\text{Bootstrap Gaussian Approx.}} + \underbrace{d_K\left(\mathcal{N}(0, \sigma^2), \sqrt{n}(\hat{\tau}_M^{bc} - \tau)\right)}_{\text{Main Theorem} \leq C(B_1+B_2+B_3)}$$

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- Because the multipliers W_i, V_i are independent Gaussians, the conditional bootstrap statistic is exactly distributed as $\mathcal{N}(0, \hat{\sigma}^2)$. The bound follows since

$$d_K\left(\mathcal{N}(0, \hat{\sigma}^2), \mathcal{N}(0, \sigma^2)\right) = d_K\left(\mathcal{N}(0, (\hat{\sigma}/\sigma)^2), \mathcal{N}(0, 1)\right) \leq \frac{\sigma}{\hat{\sigma}} \vee \frac{\hat{\sigma}}{\sigma} - 1 \stackrel{\text{w.h.p.}}{\leq} \mathcal{E}(\mu, \hat{\mu}).$$

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- A robust alternative is measuring covariate similarity using distances between component-wise ranks. [Rosenbaum '05, '10]
- **The Rank-Based Estimator:** Let $\hat{F}_n(x) := (\hat{F}_{n,1}(x_1), \dots, \hat{F}_{n,m}(x_m))^T$ be the marginal empirical CDF vector. Denote the rank-transformed covariates $\hat{L}_i := \hat{F}_n(X_i)$.
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The rank-based estimator $\hat{\tau}_{r,M}^{bc}$ is defined by considering M -NN matches over $\{\hat{L}_i\}_{i=1}^n$.
- Our non-asymptotic bounds carry over. We establish explicit finite-sample Gaussian approximation bounds for $\hat{\tau}_{r,M}^{bc}$ and construct valid multiplier bootstrap CIs.

Concluding Remarks

- **Non-asymptotic bounds:** We establish explicit finite-sample Gaussian approximation bounds for the bias-corrected covariate/rank-based matching ATE estimator.
- **Relaxed Hölder densities:** We extend the group-specific conditional covariate densities from standard Lipschitz continuity to Hölder continuity.
- **Multiplier bootstrap:** We develop a multiplier bootstrap procedure, which allows construction of fully data-driven, non-asymptotic confidence intervals accommodating a diverging M .

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Thank you for your attention!
